

Project Summary: Turn10

A Relaxing 1010-Style Mobile Puzzle Game

Introduction

Turn10 is an engaging, mobile-first puzzle game heavily inspired by the popular mechanics of "1010!". The game challenges players to strategically place varying geometric shapes onto a grid. The core objective is to form complete rows or columns, which then clear from the board, granting points and freeing up space for subsequent moves. Turn10 distinguishes itself from standard block puzzles through its signature shape rotation mechanic, simple aesthetics, and a heavy emphasis on a calming, "Zen-like" atmosphere.

Core Gameplay & Mechanics

At its heart, Turn10 relies on spatial awareness and forward-thinking strategy:

- **The Grid System:** The game utilizes a robust grid architecture where players interact with dynamically generated shapes.
- **Shape Rotation (The Hook):** Unlike traditional 1010 games, players have the ability to rotate the shapes before placing them. This unique hook opens up entirely new strategic possibilities and board management techniques.
- **Shape Placement:** Players drag and drop shapes from a tray onto the main grid. The game utilizes smooth drag-and-drop mechanics to ensure a highly responsive mobile experience.
- **Line Clearing & Scoring:** Filling a vertical or horizontal line clears the blocks. Multiline clears trigger special "Tranquility Bonuses", rewarding the player for setting up complex combos.
- **End State:** The game ends when there is no longer enough open space on the grid to place any of the currently available shapes in the tray.

Singular Zen Experience

To maximize the game's relaxing nature, Turn10 focuses entirely on a singular **Zen Mode**. By eschewing stressful timers, harsh penalties, or fragmented rule sets, the game provides a continuous, stress-relieving puzzle environment where players can progress and strategize at their own pace.

Aesthetics & "Juice"

A major pillar of Turn10 is its presentation. The game is designed to be a clean, sensory-rich experience:

- **Visuals:** The UI prioritizes simple, elegant visuals. While styles like Glassmorphism were initially tested, they were ultimately discarded in favor of a custom "Droplet Shader." This shader gives the game a cleaner, premium, and highly tactile feel without relying on cluttered visual themes.

- **Audio Design:** The calming soundscape features bespoke, AI-generated relaxing musical tracks. Alongside this, satisfying and snappy sound effects for block placement and multi-line clears were crafted using the **jsfxr** synthesizer, guided by AI parameter suggestions.
- **Game Feel (Juice):** To enhance the tactile feel of the puzzle pieces, interactions with the shapes utilize subtle DOTween animations. Combined with the relaxing background music and responsive sound effects, this creates a rewarding, satisfying sense of “juice” without disrupting the calm atmosphere.

Technical & Development Foundation

Turn10 was developed using the **Unity Engine** (C#) in a fast-paced, game jam-like environment. Notably, the project was built utilizing the **Antigravity IDE**, heavily leveraging AI agents to accelerate development. The resulting architecture demonstrates strong software design principles suited for mobile deployment:

- **Modular Systems:** Gameplay logic is cleanly divided into core managers (GridSystem, ScoreManager, ShapeManager, VisualManager).
- **Data-Driven Design:** Heavy use of ScriptableObjects for *Shape Databases* and *Game Mode Settings*, allowing designers to easily tweak balance and add new shapes without touching code.
- **Modern UI & Input:** Utilizes Unity’s new Input System for precise touch controls, and TextMesh-Pro for crisp, scalable typography across all mobile resolutions.
- **Third-Party Integration:** Relies on DOTween for highly optimized, code-driven animations and tweening.

Post-Mortem & Future Iterations

A critical part of the Turn10 development process was evaluating what mechanics served the core vision and what needed to be cut. Key takeaways from this project include:

- **Architectural Evolution (Events over Singletons):** While the initial manager-based architecture allowed for rapid prototyping, relying on numerous Singletons created tight coupling. A key focus for future iterations is implementing an Event Bus pattern to handle communication, drastically decoupling systems and improving scalability.
- **Strict Adherence to SOLID Principles:** A major takeaway is the importance of the Single Responsibility Principle—moving away from massive “God Objects” or bloated managers. Future development will prioritize ensuring that a single script handles only a single, isolated logic component.
- **Subtractive Design (Less is More):** Early builds included arcade-style floating text and complex Glassmorphism shaders. Playtesting revealed that these elements detracted from the relaxing experience. Pivoting to a minimalist Droplet shader and cutting the floating text was a crucial lesson in stripping away features to strengthen the core identity.
- **Collaborative AI Workflow (Antigravity IDE):** The development process highlighted the strengths of working iteratively alongside AI agents. While the Antigravity IDE rapidly built out the scene hierarchy and logic structures, the actual Prefab creation and component linking were done manually in tandem. This collaborative “co-pilot” approach—where the AI scaffolds the archi-

tecture and the developer refines and connects the Prefabs—proved incredibly efficient and easy to manage, setting a standard for future workflows.